

Fixed-Point Grasping with the DJI RoboMaster EP

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Abstract—This report documents a five-cycle fixed-point pick-and-place experiment using the arm and gripper mounted on a DJI RoboMaster EP. A ROS 2 Humble and Gazebo Fortress simulation produced five successful placements and five verified returns home, with a mean horizontal error of 4.358 mm and a maximum of 5.604 mm. The group also observed five successful physical transfers without an abnormal event in the nominal trials. The video’s final empty grasp is a separate, deliberate object-removal test, as clarified by the team. The nominal 5/5 result is team-confirmed; no instrumented hardware accuracy, force, payload, or timing data are claimed. The report also states the interface differences introduced by replacing the unavailable standalone mechArm 270 with the RoboMaster platform.

Keywords—fixed-point grasping, RoboMaster EP, ROS 2, Gazebo, pick-and-place, repeatability

1. SCOPE AND ORGANIZATION

1.1. Objective and platform adaptation

The required task is fixed-point pick-and-place without visual localization: return home, approach point A, descend, grasp, lift, move to point B, release, and return home. Acceptance requires at least four successful transfers in five consecutive trials, together with collision, joint-limit, and failure checks.

Because the specified standalone mechArm 270 was unavailable, the group used the arm and gripper mounted on a DJI RoboMaster EP chassis. Simulation uses ROS 2 Humble, Gazebo Fortress, and custom kinematics/control. The physical test uses the RoboMaster Python SDK. This is a documented platform adaptation, not proof of an unchanged ROS 2 hardware backend.

Information

Simulation completed 5/5 placements and 5/5 verified returns home. The group confirmed 5/5 physical transfers on 12 September 2026. Numerical accuracy is available for simulation only; the physical result is observational.

2. SYSTEM PRINCIPLE

2.1. Common task intent

Both implementations follow the same functional sequence even though their control interfaces and cycle boundaries differ.

2.2. Simulation kinematics and feedback

The simulated arm is represented by independent variables r and l , with linked joint coordinates

$$q_{\text{arm1}} = r, \quad q_{\text{arm2}} = l - r, \quad q_{\text{endpoint}} = -l, \quad q_{\text{rod}} = l. \quad (1)$$

The inverse calculation minimizes fingertip-centre error in the model’s x - z plane. It rejects residuals above 0.002 m and solutions outside the saved joint and coupling limits. Joint transitions use the quintic blend $s(u) = 10u^3 - 15u^4 + 6u^5$ to avoid discontinuous reference velocity.

Fresh bilateral gripper contact, retained-object height, and placement conditions gate later phases. Placement requires horizontal error below 23 mm, height error below 8 mm, and table contact. Home acceptance separately checks heading, translation, and arm error over a stable interval.

2.3. Physical workflow

The final script initializes the RoboMaster in AP mode and commands the arm, gripper, and chassis through the Python SDK. After each lift it stops the chassis, clears previously buffered terminal input, and waits up to 30 s for a fresh human confirmation that the object is retained. Only then is rotation allowed. Odd cycles turn $+32^\circ$ from A to B; even cycles turn -32° from B to A. After cycle 5, the remaining heading is restored before the arm moves through HIGH to HOME.

The confirmation is a supervised safety checkpoint, not a force or vision measurement. Cleanup pauses the gripper, commands zero wheel speed, and closes the SDK connection; it is not described as a certified emergency-stop system.

3. SIMULATION DEVELOPMENT

3.1. Scene and execution

The scene contains a 1.6 x 1.2 x 0.05 m table, a 25 mm cube, and fixed points $A = (0.278, -0.001, 0.0725)$ m and $B = (0.237, 0.1453, 0.0725)$ m. The cube’s simulation mass is 50 g; this is not a measurement of the real object. The launch file starts Gazebo, the bridge, model spawning, state feedback, and the task controller. Community model resources are attributed to [jeguzzi/robomaster_ros](#), archived at commit c05a39d7f0fa.

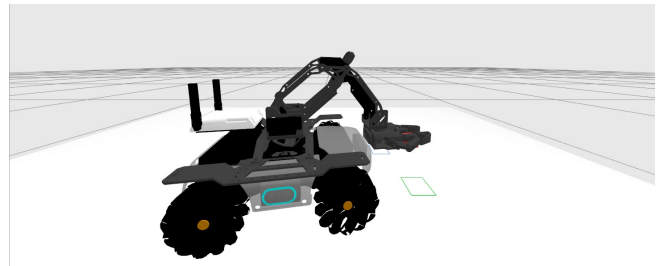


Figure 2. Recorded simulation frame during a lifted-object phase.

3.2. Five-cycle result

Table 2. Main recorded simulation run.

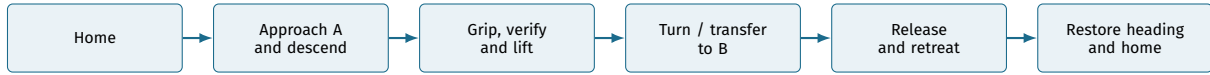
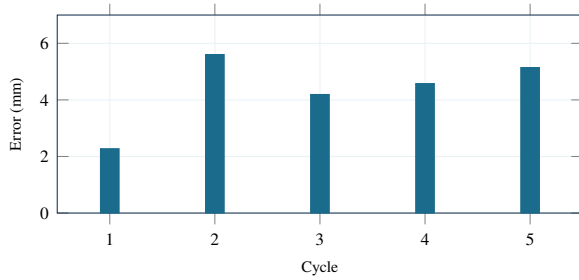
n	Err. (mm)	Head. ($^\circ$)	Time (s)	Place	Home
1	2.274	0.445	67.01	Yes	Yes
2	5.604	0.329	59.72	Yes	Yes
3	4.192	0.404	60.92	Yes	Yes
4	4.577	0.459	61.09	Yes	Yes
5	5.144	0.405	61.29	Yes	Yes

Mean horizontal error was 4.358 mm and the maximum was 5.604 mm. The five cycle intervals totalled 310.03 s; the run from initial home entry to final completion lasted 318.02 s. The trajectory contains 4475 rows over a longer span because it includes idle recording, so row span is not used as task time.

Table 1. Team organization and principal responsibilities.

Member	Area	Main responsibility
Bo Ning	Simulation	Environment, scene, models, resources, and launch setup.
Junbiao Huang	Simulation	Motion program, feedback logic, and repeated-cycle debugging.
Yongyue Qi	Analysis	Result/log audit, error analysis, report materials, and repository integration.
Yutong Xue	Hardware	Pick/release waypoints, approach/lift height, and gripper-retention checks.
Lintai Bao	Hardware	SDK communication, task sequence, chassis turn/return, and exception handling.

Xue and Bao jointly performed physical setup, parameter review, and integrated trials. All five members share the final project outcome.

**Figure 1.** Shared fixed-point grasping workflow.**Figure 3.** Horizontal placement error by cycle.

4. HARDWARE VALIDATION

4.1. Final physical parameters

Table 3. Arm waypoints used in the physical workflow.

Point	x (mm)	y (mm)	Function
HOME	82	80	Retracted high
HIGH	191	80	Extended high
HOVER	191	-1	Approach / lift
GRASP	191	-51	Pick / release

The arm descends from HOVER to GRASP, closes the gripper at SDK power 100, lifts 50 mm, and waits for human confirmation. The physical workflow alternates A→B and B→A because it does not automatically replenish the object. Simulation instead uses a refill node after home verification for cycles 1–4 only.

4.2. Observed outcome

Table 4. Team-observed physical transfers.

n	Direction	Transfer	Abnormality
1	A→B	Successful	None observed
2	B→A	Successful	None observed
3	A→B	Successful	None observed
4	B→A	Successful	None observed
5	A→B	Successful	None observed

All five lifts were confirmed. This is a team-observed record, not a sensor-generated hardware log.

4.3. Separate video evidence

The real-device video is submitted as a separate artifact and is not embedded in this PDF. Its final segment documents an additional deliberate object-removal test, excluded from the five nominal transfers. At approximately 58–61 s an operator removes the target; at approximately 61–69 s the empty gripper closes and lifts. No subsequent transfer rotation is

visible before the clip ends. The test intent and nominal 5/5 result were clarified by the team. The clip does not provide a legible terminal transcript or an instrumented stop-time record.

Table 5. Hardware-video metadata.

Field	Recorded value
Filename	RoboMaster_EP_Hardware_Demonstration.mp4
Duration	74.87 s
Frame	1280 x 720 at 29.97 fps
Encoding	HEVC Main; AAC LC stereo
Size	8,024,795 bytes
SHA-256	614a8eb55d8883cce4494496a199c828 bb679b693f58a325319570a89fad6a6e

5. PROBLEMS AND SOLUTIONS

5.1. Resources and reference frames

Early model problems included unresolved mesh paths and inconsistent end-effector references. The archive bundles the model dependency and provides `prepare_paths.py` to construct a runtime copy using local mesh paths. Kinematics uses the forward ends of gripper link 4 rather than a rear linkage, separating resource-resolution failures from control failures.

5.2. Unstable lift and false closure

A close command does not prove object retention. Simulation combines fresh bilateral contact, a small closure preload, a hold interval, and an object-height check. Hardware uses a 50 mm lift and fresh human confirmation. Neither proves gripping force, but both prevent command completion from being misclassified as a successful grasp.

5.3. Repeated-cycle state

Earlier sequences could drift or stop after one cycle. The final simulation records five placement checks, five home checks, and four refill acknowledgments, then ends after cycle 5. Hardware alternates directions and compensates the remaining heading after cycle 5.

5.4. Abnormal-case evidence

Archived notes record eight passing offline tests across model and five-cycle checks. An unreachable target such as (10, 10, 0.6) is rejected by the solver with a `ValueError`; joint and coupling limits are checked before motion. Runtime stale-feedback and phase-timeout guards also exist.

Hardware case H-01: object removed before grasping. The team deliberately removed the target to exercise the grasp-failure path. The visible empty lift and absence of subsequent transfer are consistent with holding at the supervised confirmation checkpoint. In the archived SDK script, only a fresh `y` authorizes transfer. A negative/blank response or

Table 6. Requirements, deviations, and evidence status.

Handout expectation	Evidence status
mechArm 270 hardware	Replaced by RoboMaster EP because of equipment availability.
Same ROS 2 interface for hardware	Not demonstrated; hardware uses the RoboMaster Python SDK.
Parameter-only migration	Not demonstrated across the different control backends.
Each member's individual safety check	No separate individual check record was supplied.
At least 4/5 transfers	Met in the recorded simulation and team-observed hardware outcome.

Published Git history RoboMaster EP Experiment 2		
Source: git log -14 --format="%h %cs %s"		
Local data rendering; not a GitHub webpage screenshot. Snapshot: c232dcd		
c232dcd	2026-09-14	docs(evidence): distinguish nominal success from deliberate empty-grasp test
3f6cd65	2026-09-13	chore: remove individual-report artifacts
2ae9bc8	2026-09-13	docs(group): redesign report with Tau lab template
149fd0b	2026-09-13	docs(group): add brief experiment report and final evidence index
4306259	2026-09-13	media(hardware): add real-device demonstration video
dcc24e3	2026-09-13	docs(reports): remove placeholder-like wording

Figure 4. Commit-history evidence exported from the repository at revision c232dcd. The package also includes the text export; a later commit publishes this revised report.

30 s timeout aborts above the source without rotation or homing. The finally block attempts gripper pause, zero wheel speed, and SDK closure. These are code-defined actions; the exact input, timeout branch, and successful cleanup calls are not independently established by the video. H-01 is an additional negative test, not a failed nominal trial or an automatic object detector.

The accompanying abnormal-test record separates observation from expected behavior. Communication loss, runtime limit violations, stale feedback and refill failures were not demonstrated as injected system faults. No separate individual safety-check record was supplied.

6. REPOSITORY TRACEABILITY

The public shared repository is github.com/ccx8kg5qpd-debug/robomaster-ep-fixed-point-grasping. The experiment was completed before repository assembly. The history records the genuine review, organization, publication, and reporting sequence on 13–14 September 2026; it is not presented as a fabricated contemporaneous development history.

7. SUBMISSION AND CONCLUSION

7.1. Group package manifest

Table 7. Materials required by the handout.

Material and package location	Status
ROS 2 program, launch, and parameters: <code>01_simulation/</code>	Included
Model and point/height setup: <code>01_simulation/</code>	Included
Simulation video: <code>03_videos/simulation/</code>	Included
Real-device video: <code>03_videos/hardware/</code>	Included
Five-cycle records: <code>04_results/</code>	Included
Abnormal-test record: <code>04_results/abnormal_test_record.md</code>	Included
Brief group report: <code>05_group_report/</code>	Included
GitHub link and commit evidence: report and README	Included

7.2. Reproducibility

Simulation targets ROS 2 Humble and Gazebo Fortress on the team's Jetson Linux environment. The startup wrapper contains a workspace path that must be adapted on another machine. The model dependency

is bundled with attribution. Saved `results.json`, `events.jsonl`, and `trajectory.csv` support independent checking of the numerical results without rerunning Gazebo.

7.3. Conclusion

The evidence supports five successful simulation cycles with home verification and five team-confirmed nominal physical transfers without observed abnormality. The separate object-removal test demonstrates an empty-grasp scenario with no subsequent transfer visible; exact terminal and cleanup events remain unlogged. The project demonstrates a complete fixed-point sequence on the available RoboMaster EP. Its strongest quantitative evidence is the saved simulation record; the physical result is supported by team observation and a separate real-device video, not instrumented precision data.

7.4. Sources

- [1] Course handout, *Robot Integration Group Project I: Fixed-Point Robotic-Arm Grasping*, S16–S20 / E5.
- [2] Public team repository, main branch, revision of 14 September 2026.
- [3] Primary simulation record 20260913_153855_682865; comparison record 20260913_151835_077998.
- [4] Hardware-video metadata and SHA-256 in `media/hardware/README.md`.
- [5] jeguzzi/robomaster_ros, archived commit c05a39d7f0fa.
- [6] Guillermo Jimenez, *Tau Class v2.6.0*, CC BY 4.0.